

Ruhila Goswami

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"Nothing exists for itself alone, but only in relation to other forms of life." Charles Darwin

Research Profile

Focus: Genetics; within-population variation; evolutionary divergence.

Interests: Molecular and developmental biology; genetic control of gene expression and cell fate; regulatory pathways and cellular stress responses; population genetics; phenotype-genotype mapping; structural variation and selection; salmonid evolution; non-invasive methods; nucleic-acid delivery; reproducible computational biology.

Methods: Phylogenetic inference, genomic synteny, PCR-based sex determination and DNA isolation, skeletal phenotyping (tooth counts and bone shape), field sampling, mammalian cell culture, lentiviral packaging, and reproducible analysis in R and Python.

Education

University of Iceland 2024-2026
M.S. Biological Science, Reykjavík, Iceland
Advisor: Prof. Arnar Pálsson (co-supervisor Guðbjörg Ósk Jónsdóttir). Thesis: Evolutionary Divergence in Dental Variation and Sex Chromosome Architecture in Icelandic Brown Trout and Arctic Charr: feeding morphology (tooth numbers and craniofacial bone shape across populations and sympatric morphs) linked to ddRAD-seq population and sex-chromosome structure.

IISER Mohali 2021-2024
B.S. Biology (Major), India
First Division with Honors.

Research Experience

University of Iceland 2024-2026
M.S. research with Prof. Arnar Pálsson
BTAC morphology pipeline (six craniofacial bones, hierarchical models, allometry, PCA) and ChromSex/SexPCR/rsx stack for sex-balanced F_{ST}, multi-species *sdY* PCR, and streaming RAD-seq sex-marker discovery. Field and lab: electrofishing, dissection, otolith and bone preparation, DNA isolation, PCR sexing.

University of Iceland Summer 2023
Research Staff with Prof. Arnar Pálsson
Synteny of histone genes in salmonids: gene arrangement and neighbouring genomic regions.

University of Iceland Summer 2022
Research Staff with Prof. Arnar Pálsson
Reproducible phylogenetic workflow for molecular evolutionary pathways in salmonids (NCBI curation, BLAST, MUSCLE5, Gblocks, BIONJ/RAxML-NG on HPC). Project Report: Computational Primitives for Evolutionary Paths (≈ 147 pages).

IIT Kanpur Winter 2022
Research Interns with Prof. Debabrata Goswami and Dr. Nagma Parveen
Non-linear optics and optical tweezers; wet-lab viral methods and a SARS-CoV-2 pseudovirus protocol (HIV-1 lentiviral packaging, luciferase reporter).

IISER Mohali Summer 2021
Research Intern with Dr. Lolitika Mandal
Genetic tools for *Drosophila*: wet-lab data collection and analysis.

Manuscripts

rsx streaming toolkit 2026
Published, *BMC Bioinformatics*, doi:10.1186/s12859-026-06628-4: rsx, a high-performance streaming toolkit for RAD-seq sex determination (with Rohit Goswami). Software and Zenodo archive 10.5281/zenodo.20531539.

Tooth numbers and bone shape 2026
Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids (BTAC analysis; manuscript draft with K. A. Williams, I. Eloidin, G. Ó. Jónsdóttir, and A. Pálsson).

Brown trout LG28 / *sdY* 2026
Concordant PCR and RAD-seq evidence localises the brown trout (*Salmo trutta*) sex locus to distal LG28 across Icelandic populations (with M. Lagunas and A. Pálsson). Audited *sdY* PCR vs dissection and population-replicated RAD recovery of the distal LG28 sex

interval.

Posters, Awards, and Science Communication

Posters 2025-2026
Icelandic Ecological Society (Vistís), NOWPAS, and IceBio on tooth numbers and bone shape in Icelandic salmonids; **Best Poster Award**, Vistís 2025. BES 2022 poster on *Salmo salar* histone sequence data.

Proceedings 2022
IEEE PDGC on reproducible literate phylogenetic analysis and high-performance computing workflows (with R. Goswami and co-authors).

Vísindavefurinn June 2026
Co-author with A. Pálsson: *Hvernig þróast ósamhverfa í byggingu dýra?* (How does asymmetry evolve in animal body structure?), University of Iceland science portal.

Teaching

LÍF401G Developmental Biology Spring 2026
Teaching Assistant, University of Iceland (Prof. Adam Ray Smith); assisted with embryo analysis.

LÍF541G Plant Physiology Fall 2025
Teaching Assistant, University of Iceland (Prof. Victor Anthony Albert); led discussions in an undergraduate course focused on plant genetics.

LÍF403G Evolutionary Biology Spring 2025
Teaching Assistant, University of Iceland; evolutionary theory, natural selection, adaptation, and the history of life.

Research Skills

Genomics: Population genetics, genomic synteny, structural variation, sex-balanced F_{ST} (Hudson / Weir & Cockerham), PCR-based sex determination, non-invasive sampling.

Phenotypes and fieldwork: Dental and craniofacial variation, bone extraction, otolith extraction, fish dissection, electrofishing, salmonid sampling and preservation.

Evolutionary methods: Phylogenetic tree construction (distance, maximum likelihood, Bayesian), BLAST, MUSCLE5, Gblocks, RAxML-NG, BEAST2.

Computing and lab: R, Python, shell, Git, Snakemake, `targets`/pixi pipelines, reproducible literate programming, DNA isolation and preservation, PCR, microscopy, *Drosophila* tissue dissection, mammalian cell culture, HIV-1 lentiviral packaging.

Open tools (RuhiRG): ChromSex (sex-balanced F_{ST} and sex-locus annotation); SexPCR (multi-species *sdY* assay audit); rsx (streaming RAD-seq sex-marker discovery, with HaoZeke/rsx-rs).

Service and Affiliations

Google Summer of Code 2023-2024
Student Developer on Python bindings and documentation for d-SEAMS/pySEAMS (installation, usability, repository maintenance).

IEEE P3173 2022-present
Secretary, IEEE Standards Committee for Endocrine Disrupting Chemical Hazard Labeling.

Memberships: Nordic Society Oikos; Icelandic Ecological Society (Vistís); Icelandic Biological Society (Líffræðigáttin); British Ecological Society; Genetics Society, UK; Genetics Society of America.

Referees

Prof. Arnar Pálsson: MSc supervisor, Institute of Life and Environmental Sciences, University of Iceland. apalsson@hi.is

Prof. Victor Anthony Albert: Course supervisor (LÍF541G), University of Iceland. victoralbert@hi.is